

7 Read the passage below and answer the questions that follow.

Solar Power in Singapore

Singapore has set a target for solar energy which aims to produce enough power by 2030 to meet the annual needs of 350 000 households. The present total installed capacity is 400 MWp (mega-watt-peak is the power output of a solar power system which would be achieved under ideal conditions). This is a tiny fraction of the country's total power output of 12 600 MW produced by power stations with an overall efficiency of 27%. The power stations use natural gas as a fuel that has an energy density of 56 MJ kg^{-1} .

Solar photovoltaic (PV) panels consist of a number of cells composed of semiconducting materials that convert sunlight into electricity through what is known as the photovoltaic effect. The constraints to Singapore's ability to host a substantial solar photovoltaic capacity arise from limited availability of two natural resources - sunlight and space.

Although Singapore's climate is relatively hot and the weather is usually sunny, the average intensity of solar radiation across a full year is significantly less than that in northern China, and in the deserts of North Africa, the Middle East and Australia. The average solar intensity in Singapore is 780 W m^{-2} . The average monthly sunshine hours in Singapore in 2020 is shown in Fig. 7.1.

month	total sunshine hours / h
January	170
February	185
March	195
April	175
May	180
June	180
July	190
August	180
September	155
October	155
November	130
December	135

Fig. 7.1 (Source: Meteorological Service Singapore)

Space is the second key constraint. Singapore lacks vast open spaces to build solar arrays. Today, about one third of the country's solar energy capacity sits on the rooves of residential buildings. A detailed analysis produced by a consortium led by the Solar Energy Research Institute of Singapore (SERIS) concluded that the total usable space for solar PV panels amounted to just 37 km^2 , out of the total land area of 728 km^2 in Singapore.

The maximum efficiency of commercially available PV cells currently is 20 %. The life span of well-maintained solar panels is 25 years. This compares to 40 years for a gas-fired power station. The panels will then need to be replaced and the materials recycled.

- (a) (i) Use the unit MJ kg^{-1} to deduce what is meant by *energy density*.

.....
 [1]

- (ii) Determine the rate of consumption of natural gas in the power stations.

rate = kg s^{-1} [2]

- (b) The radiant power of the Sun is $3.90 \times 10^{26} \text{ W}$. The average radius of the Earth's orbit about the Sun is $1.50 \times 10^{11} \text{ m}$.

- (i) Calculate the solar intensity incident on Earth.

intensity = W m^{-2} [2]

- (ii) State an assumption you have made in (b)(i).

.....
 [1]

(iii) Suggest two reasons for the difference between the average solar intensity in Singapore and your answer in (b)(i).

1.
.....
2.
..... [2]

(c) The symbol for a PV cell is shown in Fig. 7.2.

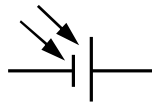


Fig. 7.2

To provide a useful supply, many identical PV cells are connected in a series and parallel array, as shown in Fig. 7.3.

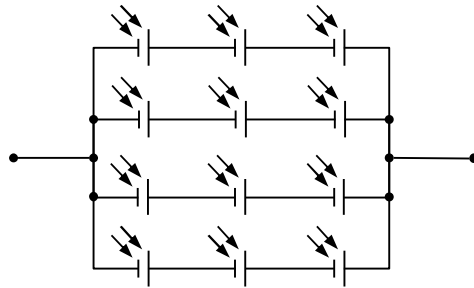


Fig. 7.3

Explain

(i) one advantage of connecting the cells in series,

-
..... [1]

(ii) two advantages of connecting the cells in parallel.

1.
.....
2.
..... [2]

- (d) Use appropriate values from the passage to determine the ratio

$$\frac{\text{solar power output from } 37 \text{ km}^2 \text{ of PV panels}}{\text{total power output from power stations}}$$

ratio = [2]

- (e) A PV panel of area 1.50 m^2 is installed to produce solar power for a household in Singapore. The average monthly household electricity consumption is 480 kWh.

The electricity tariff is \$0.23 per kWh.

- (i) Calculate

1. the solar power produced by the PV panel,

power = W [2]

2. the electrical energy produced by the PV panel in 2020.

energy = kWh [2]

- (ii) The cost of installing a PV panel is \$750.

Using appropriate values from the passage and your answer in (e)(i), write an argument for or against the use of solar power. You should show calculations with clear working in support of your argument.

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..... [3]

End of paper